

Lecture 11

Experimental Design and Research Ethics

Review

1 What is the difference between an observational study and an experiment?

An experiment assigns the treatments; an observational study only watches and measures. The test is who decided which group each unit ended up in.

2 What is a lurking variable?

A variable associated with both the explanatory and the response variable, which can make two things look related when neither causes the other.

3 What does it mean for a study to be blinded, and what makes it double blind?

Blinded: the subjects do not know which treatment they received. Double blind: the people measuring the response do not know either.

A placebo surgery

Arthroscopic knee surgery for osteoarthritis was one of the most common operations in the United States, and patients said it helped.

Moseley and colleagues (2002) enrolled 180 patients with knee osteoarthritis at the Houston VA Medical Center and assigned each of them at random to one of three groups: arthroscopic debridement, arthroscopic lavage, or placebo surgery - skin incisions and a simulated procedure, with no arthroscope inserted.

Neither the patients nor the staff who later measured their pain and function knew which group a patient was in. Every patient was followed for two years.

What kind of study is this, and what is it about the design that lets it answer the question?

A completely randomized, double blind experiment.

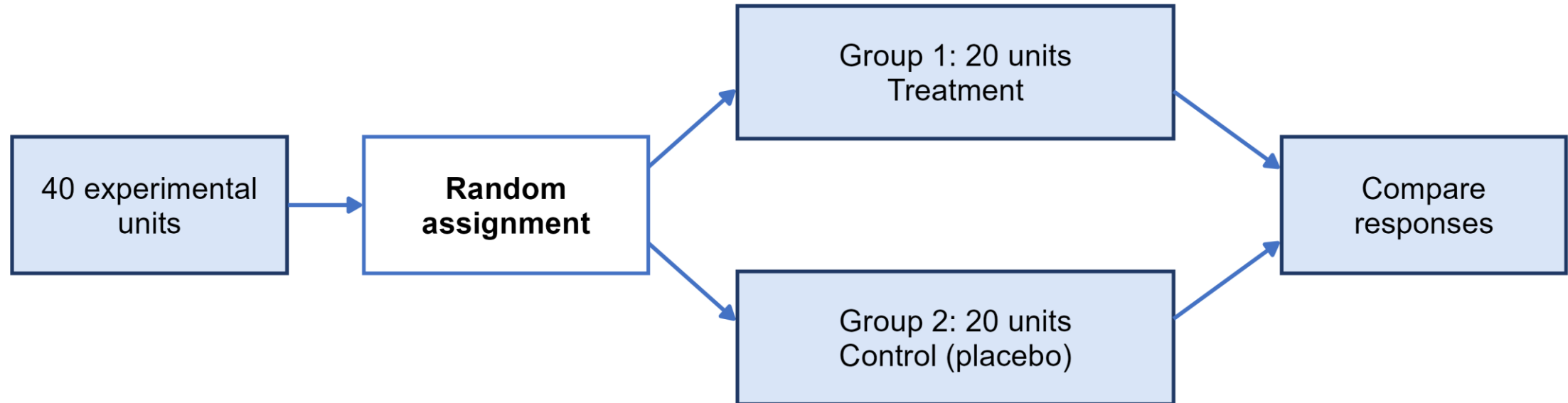
Three treatments assigned at random, one of them a placebo; double blind because both the patients and the assessors were kept unaware of who got which. Over two years neither surgery group did better than the placebo group on pain or function - the benefit patients felt came from believing they had been treated. Without a placebo arm and blinding, nobody would have known.

Source: Moseley et al. (2002), New England Journal of Medicine

Completely randomized design

completely randomized design

every unit is assigned to a treatment completely at random - the simplest design



The Pfizer-BioNTech vaccine trial was a completely randomized design with two groups

Randomized block design

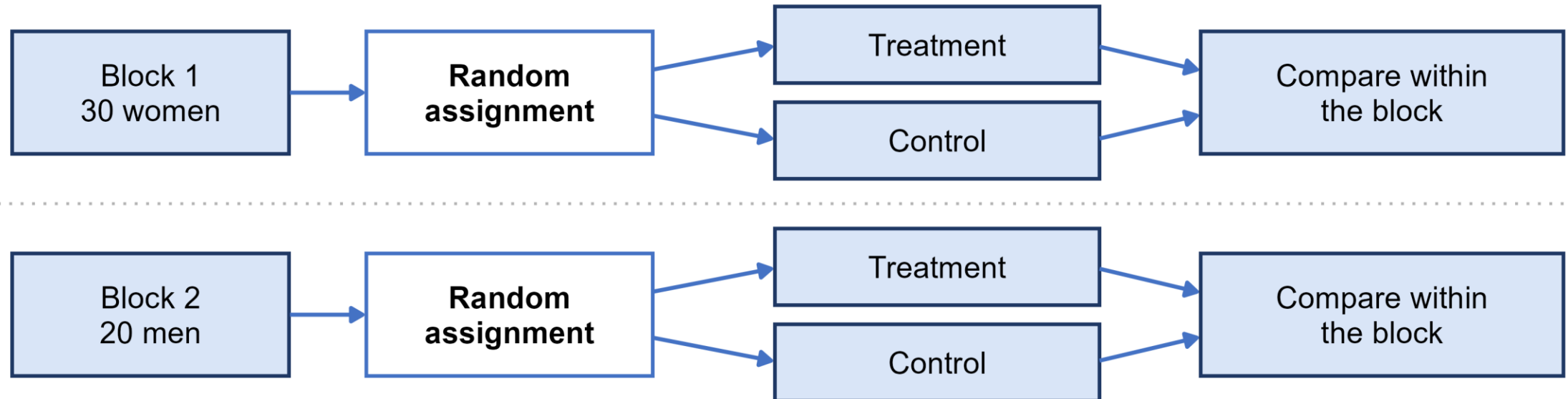
block

a group of units that are alike in a way that affects the response, formed BEFORE treatments are assigned

randomized block design

randomize treatments separately inside each block, then compare treatments within blocks

blocking is to experiments what stratifying is to sampling

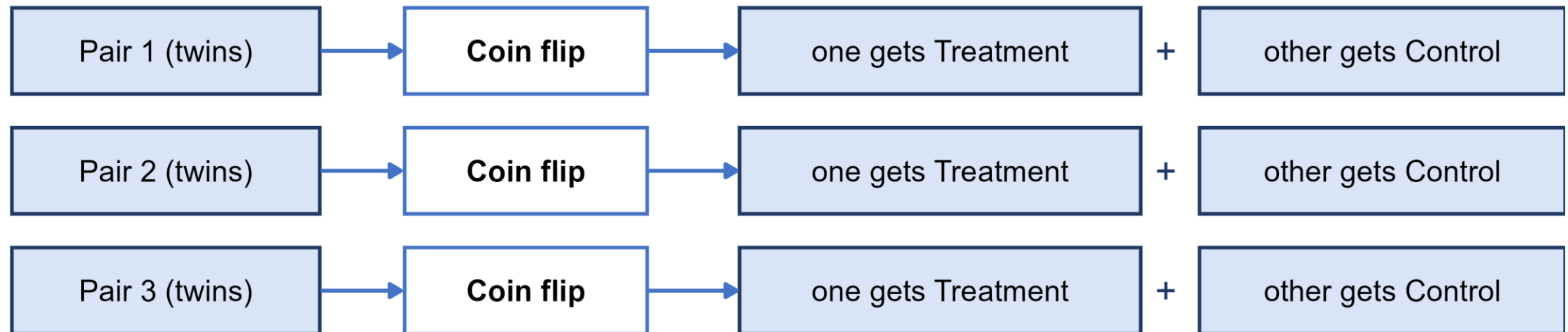


Matched pairs design

matched pairs design

a block design where every block is a pair of very similar units, and a coin flip decides which one gets each treatment

or each subject is their own pair, receiving both treatments in a random order



Before-and-after data with no random order or control group is much weaker evidence

What does randomization buy you?

Random sampling lets you generalize from the sample to the population

Random assignment lets you conclude cause and effect

Most studies have only one. A trial on volunteers shows a vaccine works - for people like those volunteers

		Randomly ASSIGNED to treatments?	
		Yes	No
Randomly SAMPLED?	Yes	Cause, AND generalize to the population rare: an experiment on a random sample	Generalize, but not cause e.g. a random-sample survey
	No	Cause, but only for units like these e.g. most clinical trials, which use volunteers	Association in these units only e.g. volunteers who chose their own treatment

Was the placebo surgery trial ethical?

Patients in the placebo arm were anaesthetised and cut open with no chance of benefit. How was that allowed?

Every patient signed a consent form stating explicitly that they might receive placebo surgery and that it would do nothing for their arthritis, and an institutional review board approved the trial.

Why was it acceptable to withhold an operation that was already standard care?

Because nobody actually knew it worked. It was being performed very widely on the strength of what patients reported, with no randomized evidence behind it. When experts are genuinely uncertain, a trial is justified - that uncertainty is called equipoise.

So, was it ethical?

On balance most would say yes: real informed consent, board approval, genuine uncertainty, and a small risk set against an answer that changed practice for hundreds of thousands of people. Sham surgery is still debated, and every such trial is judged case by case.

Ethics: the Facebook experiment

For one week in January 2012, Facebook ran a completely randomized experiment on 689,003 users: each was assigned at random to a News Feed with fewer positive posts, with fewer negative posts, or left unchanged as a control. The results were published in 2014. The users never knew.

1. Facebook said agreeing to its Data Use Policy counted as informed consent. Do you agree?

The journal did not: its Expression of Concern said the study may not have met the principles of informed consent and the chance to opt out.

2. Why did no ethics board stop it?

Facebook, a private company, was not bound by the federal rules. Cornell's board decided its researcher was not directly involved, so it did no review.

3. What is the lesson?

Clicking I agree on a terms-of-service page is not informed consent - even for a well-designed experiment.

Sources: Kramer, Guillory and Hancock (2014), PNAS; PNAS Editorial Expression of Concern (2014)

The Belmont Report: three principles

Why the rules exist: in the U.S. Public Health Service syphilis study at Tuskegee (1932 to 1972), hundreds of Black men with syphilis were left untreated without informed consent. The Belmont Report (1979) followed.

respect for persons

people decide for themselves whether to take part

beneficence

do no harm: maximize the possible benefits and minimize the possible risks

justice

the burdens and benefits of research are shared fairly

Research ethics today

informed consent

subjects are told the purpose, procedures and risks in advance, and can refuse or quit at any time

confidentiality

information about individuals is protected and never released in a form that identifies them

Institutional Review Board (IRB)

reviews and approves studies on human subjects before they start
at UI, through the Office of Research Assurances

Ethics: honest data

Ethics also means honest data. An investigation of psychologist Diederik Stapel, infamous social psychologist, found he had invented data: it tainted more than 55 of his papers and 10 PhD dissertations he supervised. A researcher is surveying households in a community:

Source: Introductory Statistics 2e (OpenStax), Section 1.4 and Example 1.22

Practice: what design, and can it show cause?

1. A survey finds that people who own dogs have lower blood pressure than people who do not.
Observational - nobody was assigned a dog. Active people may both own dogs and have lower blood pressure.
2. 60 volunteers with insomnia are randomly split: 30 get melatonin, 30 get an identical-looking placebo.
Completely randomized experiment - it can show cause, for people like these volunteers.
3. In each of 4 greenhouses, tomato plants are randomly assigned to fertilizer A or fertilizer B.
Randomized block design, with greenhouses as the blocks - it can show cause.

What you should know: studies and designs

Three kinds of study - be ready to define and identify each one

An experimental study assigns the treatments. An observational study only watches and measures. A survey samples a population in order to describe it, and assigns nothing.

The sampling designs

Simple random, systematic, stratified, cluster, and two-stage cluster. Convenience and voluntary response samples are not random, and they are biased.

The experimental designs

Completely randomized, randomized block, and matched pairs.

What you should know: terminology

The language of an experiment

Experimental units, the factor, its levels, the treatments, the control group, a placebo, blinding and double blinding.

Association is not causation

A lurking variable is associated with both the explanatory and the response variable, and can produce an association where there is no causal link.

The two things randomness buys you

Random sampling lets you generalize to the population. Random assignment lets you claim cause. A bigger sample shrinks sampling error, but it never removes bias.